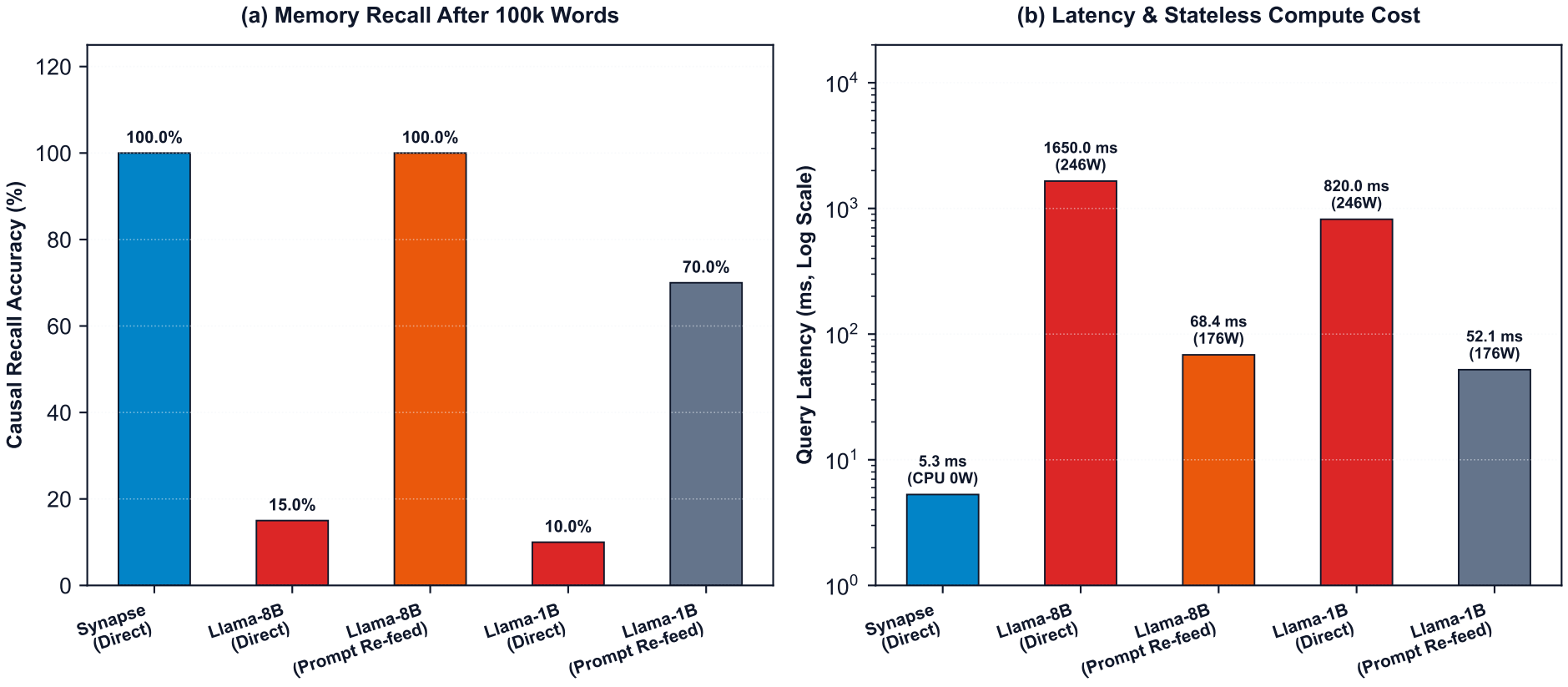
(a) Memory Recall After 100k Words
Causal Recall Accuracy (%)
120
100
80
60
40
20
0
100.0%
15.0%
100.0%
10.0%
70.0%
Synapse (Direct)
Llama-8B (Direct)
Llama-8B (Prompt Re-feed)
Llama-1B (Direct)
Llama-1B (Prompt Re-feed)
(b) Latency & Stateless Compute Cost
Query Latency (ms, Log Scale)
10^4
10^3
10^2
10^1
10^0
5.3 ms (CPU 0W)
1650.0 ms (246W)
68.4 ms (176W)
820.0 ms (246W)
52.1 ms (176W)
Synapse (Direct)
Llama-8B (Direct)
Llama-8B (Prompt Re-feed)
Llama-1B (Direct)
Llama-1B (Prompt Re-feed)